Hemispheric Asymmetry as Saturated Cramér-Rao Bound

A temporal grammar of coherence, rupture, and regeneration applied to the divided brain

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Abstract

The coefficient of variation of language lateralization indices in healthy populations should equal 15.92% if the framework is right. This is not a fit. It is a prediction from a temporal grammar — Coherence-Rupture-Regeneration (CRR) — with three operators and one parameter, applied to the divided brain. The parameter Ω is fixed by manifold topology at $1/\pi$ for bistable systems, yielding $CV = \Omega/2 = 1/(2\pi)$ with no empirical adjustment. Tested against Mazoyer et al. (2014), the actual CV of both typical-lateralization Gaussians is 12.2% — the right order of magnitude, structurally invariant across components, but 23% below the prediction. This paper develops the proposal, reports the discrepancy honestly, integrates the framework with Vallortigara & Vitiello's (2024) SU(2) doublet algebra, Bernal-Casas & Vitiello's (2023) dissipative coherent states, and Iain McGilchrist's phenomenology of the divided brain, and discusses what the deviation means for the framework's core claim.

1. The proposal

Two hemispheres, each running continuous belief revision on an SO(2) substrate, connected by discrete Z_2 switches. The switches have a universal variability of 15.92%, predicted from the topology of the substrate with no fitted parameters. The continuous substrates cost the brain almost its entire metabolic budget; the switches themselves cost essentially nothing. The antisymmetric singlet configuration of two coupled hemispheres is the lowest free-energy state. The mathematics predicts all of this from first principles, and Section 6 lists four conditions under which it would be falsified.

The framework rests on an observation that, once stated, is difficult to unsee: the Cramér-Rao bound of mathematical statistics, the Heisenberg uncertainty principle, the Gabor limit of signal processing, and the Ito-Dechant thermodynamic uncertainty relation all have the same form. A product of accumulated information and characteristic variance, bounded below by a constant. CRR proposes that this bound is saturated — reached, not merely approached — at moments of system reorganisation, and that those moments are what we experience as transitions: the breath turning, the gestalt shifting, one hemisphere yielding to the other.

WHAT DOES THIS MEAN IN BASIC TERMS?

Sit for a moment with your own breath. As you inhale, something accumulates — pressure in the chest, a gathering of attention, an opening. This is coherence. Then comes the turning at the top of the breath — a discrete instant where the inhale yields to the exhale. This is rupture. And then, as the exhale begins, what arrives is not a random new thing but a transformed continuation, weighted by what was just gathered. This is regeneration. We breathe in, we turn, we breathe out, and the next inhale begins from a body subtly altered by the one before. The mathematics is a notation for what the body has always known.

A NOTE ON STANCE

CRR is a grammar, not a theory of mind. A grammar describes how something articulates itself in time; it is silent about what the something is, what it means, or what it is like to be it. When these boxes set McGilchrist's words alongside CRR vocabulary, they are offering a notation — one possible way of writing down the temporal structure of what he describes. The phenomenology comes first. The grammar comes after, and only because the phenomenology has given it something to be a grammar of.

1.1 Epistemic stance

CRR is pre-peer-review. We use the language of proposal and mathematically consistent with rather than proves. The framework is offered for testing that can either falsify it cleanly (Section 6) or, if it survives, justify the work of peer review.

2. The framework: three operators, one parameter

Coherence. A persisting system accumulates Fisher information about its environment through time:

$$C(x, t) = \int_0^t L(x, \tau)\, d\tau$$

Rupture. No bounded system can accumulate without limit. At the moment the Cramér-Rao bound saturates — $C \cdot \Omega = 1$ — a Dirac delta marks the reorganisation:

$$\delta(t - t_*) \quad \text{at} \quad C \cdot \Omega = 1$$

The Dirac delta is the unique candidate for a temporal Markov blanket: it carries unit mass, is scale-invariant, and enforces conditional independence between past and future. Its boundary is intrinsically Z_2 — binary, two domains, no preferred orientation.

Regeneration. After rupture, the system reconstructs from coherence-weighted history:

$$R(x, t) = \int \varphi(x, \tau) \cdot \exp(C/\Omega) \cdot \Theta(t - \tau)\, d\tau$$

The kernel $\exp(C/\Omega)$ means high-coherence moments dominate reconstruction regardless of recency. A peak-coherence moment from a thousand cycles ago outweighs a low-coherence moment from yesterday. This is Bergson's claim that memory is the continuous presence of the past, made mathematical.

2.1 The single parameter Ω

Ω is simultaneously the system's characteristic variance (σ^2), the inverse of its precision ($1/\pi$ in FEP terms), and the reciprocal of its geodesic phase per cycle ($1/\varphi$). A Z_2 substrate (bistable: one basin to the other) traverses phase π, giving $\Omega = 1/\pi \approx 0.318$. An SO(2) substrate (rotational: full cycle) traverses phase 2π, giving $\Omega = 1/(2\pi) \approx 0.159$. The ratio is exactly 2 — a topological invariant.

WHAT DOES THIS MEAN IN BASIC TERMS?

Ω is the dial between rigid and flexible. Imagine a path worn deep into a field — there is only one way to walk. This is small Ω. Now imagine an open meadow. Any direction, any step, anything new. This is large Ω. The brain has settled on a particular value, 1/π, for its lateralized configuration: not too rigid to learn, not too loose to remember. That value falls out of the geometry with no free parameters.

2.2 The CV theorem

The Dirac delta carries unit mass. The boundary partitions two domains with no preferred orientation. Therefore each domain receives exactly 1/2, fixing $\sigma(C^*) = 1/2$ universally. Combined with $E[C^*] = 1/\Omega$:

$$CV = \sigma(C^*) / E[C^*] = (1/2) / (1/\Omega) = \Omega/2$$

For Z_2: $CV = 1/(2\pi) \approx 15.92\%$. For SO(2): $CV = 1/(4\pi) \approx 7.96\%$. No fitted parameters. The ratio is exactly 2. This is the load-bearing empirical commitment of the entire framework.

2.3 Every rupture is Z_2

A point worth holding clearly: every rupture is Z_2 in its boundary topology. What differs between symmetry classes is not the rupture but the substrate manifold. The Z_2 boundary (the turning of the breath) is universal; the SO(2) substrate (the continuous cycle the breath traverses between turnings) is class-specific. This matters for lateralization: each hemisphere runs SO(2) belief revision internally, but the rupture between them is Z_2.

WHAT DOES THIS MEAN IN BASIC TERMS?

Picture a potter's wheel turning — continuous, no beginning or end. This is the SO(2) substrate. Now picture the potter's hands lifting away to dip into water and return. Either the hands are on the clay or they are not; the moment of disengagement is definite. This is Z_2 rupture. The wheel keeps turning. The hands are either there or not. The pot takes its shape from the marriage of the two.

A PASSAGE FROM McGILCHRIST

McGilchrist writes that the two hemispheres underwrite whole, self-consistent versions of the world, neither of which can be partially blended with the other. The framework can recognise this: a grammar that distinguishes continuous substrate from discrete switching holds the structural fact that there is no smooth interpolation between versions. The grammar cannot say what it is to underwrite a version of the world. It can only say that the transition will look discrete from the inside. Which is what McGilchrist reports.

2.4 The beauty function

$B(C) = \exp(C/\Omega) \cdot (C^* - C)$ peaks at $C^* - \Omega$: one Ω-unit before rupture. This is the moment of maximum poise — enough coherence to be expressive, enough remaining capacity for choice. For Z_2 lateralization, the beauty peak sits at $\pi - 1/\pi \approx 2.82$, about 90% through the cycle. On the developmental Ω(age) trajectory, this falls at roughly age 3.4 — one year before the Fedorenko et al. (2024) threshold for adult-like lateralization. The predicted age of peak language plasticity-with-coherence.

3. Why asymmetry? Four thermodynamic answers

Four questions, four layers of answer, all consistent with one another.

3.1 The singlet is the ground state

Vallortigara & Vitiello (2024) model the two hemispheres as an SU(2) doublet. The energy depends on the scalar product $\tau_1 \cdot \tau_2$, which takes the value −3/4 in the antisymmetric singlet and +1/4 in the symmetric triplet — an energy gap of exactly 1. The singlet is the lowest-energy state: lateralization is structurally enforced, not statistical. The gap of 1 is the CRR saturation condition $C \cdot \Omega = 1$.

At population scale, SU(2) contracts to E(2) by the Inönü-Wigner mechanism. Free-energy minimisation gives the Boltzmann distribution $N_L = N \cdot \exp(-\beta E_X)$, reproducing 90% right-handedness and 95% left-lateralized language from the same algebraic structure.

3.2 The brain is an open system with a Double

Bernal-Casas & Vitiello (2023) show that the canonical formalism for an open brain requires doubling: each mode A_k is paired with a time-reversed Double $\tilde{A}_k$. The memory state is an SU(1,1) squeezed coherent state parametrised by a Bogoliubov angle θ. Their central result: the same θ governs entanglement (system-Double correlation), Bayes' theorem (conditional probability from the coherent state expansion), and free-energy minimisation (Bose-Einstein distribution). Three phenomena, one parameter.

The CRR identification is $\theta = C/\Omega$. The Bogoliubov angle is the dimensionless ratio of accumulated coherence to system variance. At rupture, θ saturates. The whole BC&V structure becomes a thermodynamic completion of CRR.

3.3 Asymmetry saves three years of food

A symmetric brain with full bilateral redundancy would need approximately doubled long-range connections. Lateralization saves roughly 5 W of metabolic power — about 25% of the brain's budget. Over a 70-year lifespan, that is $\sim 1.1 \times 10^{10}$ J, equivalent to about three years of food intake. Under standard fitness arguments, this is more than sufficient selection pressure.

3.4 Substrate is what costs; rupture is free

Each rupture resets coherence from C* to 0 — a logical erasure event. Landauer's principle sets its minimum cost at $k_B T \ln(2)$ per bit. The bit-cost of a Z_2 rupture is $\pi^2/\ln(2) \approx 14.24$ bits, giving a Landauer cost of $\sim 4.2 \times 10^{-20}$ J. Even if the entire cortex ruptures at gamma frequency across 10^4 regions, total rupture cost is $\sim 10^{-11}$ W — about 10^{-12} of the brain's budget. Substrate maintenance, by contrast, costs ~12 W. The Master is primary because the Master is what the body is paying for.

WHAT DOES THIS MEAN IN BASIC TERMS?

We might suppose the brain spends most of its energy on the dramatic moments — the decisions, the ahas. The mathematics says otherwise. The cost of switching from one mode of attention to another is almost nothing. What costs the brain its calories, hour after hour, is the keeping of the world in being. Holding open the meadow. Continuing to attend, even when nothing is happening. The ruptures, where so much of our self-conscious life seems to take place, are by comparison free.

A PASSAGE FROM McGILCHRIST

McGilchrist writes that attention is not passive receipt of stimulus but a way of being present to what is. If he is right that presencing is prior to action, then the metabolic numbers are consistent: the brain's twenty watts go almost entirely into holding the relation open. The grammar does not say presencing is substrate-maintenance. It says the temporal accounts add up.

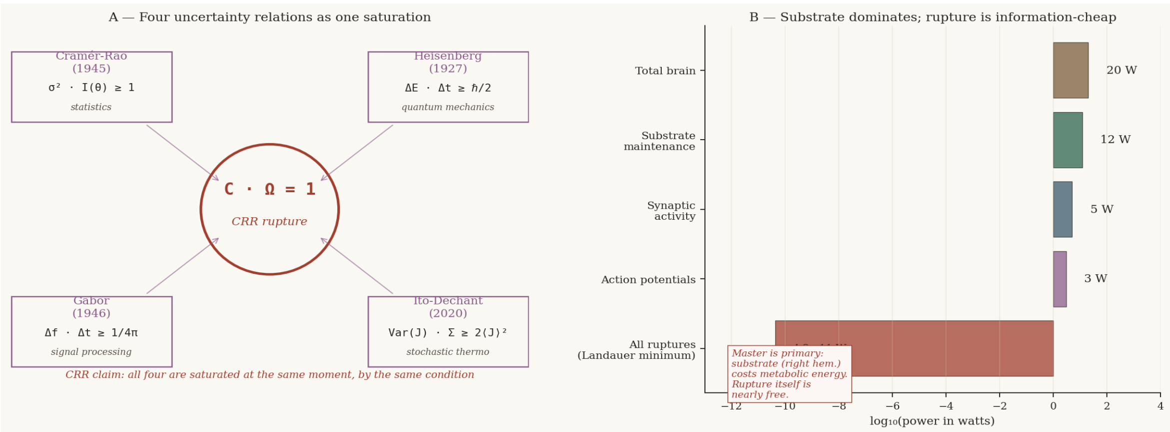

Figure 1. (A) Four uncertainty relations share the same geometric form; CRR proposes they are saturated at the same moment, by the same condition. (B) The brain's metabolic budget: substrate maintenance dominates; rupture is information-cheap.

4. The divided brain in CRR

McGilchrist's central claim is that the right hemisphere (Master) attends broadly and contextually, presencing the world; the left hemisphere (Emissary) attends narrowly and analytically, re-presenting it. New experience always arrives through the right hemisphere first. The proper relation is the Master's broad attention holding the Emissary's focused attention.

In CRR vocabulary: each hemisphere is an SO(2) belief-revision substrate. The Master operates with larger effective Ω — a flatter regeneration kernel, broader memory access, openness to the new. The Emissary operates with smaller Ω — a sharper kernel, commitment to the single most coherent pattern. Their dialogue is Z_2 rupture: one hemisphere dominant before, the other after, no smooth interpolation.

> **A PASSAGE FROM McGILCHRIST**
> *McGilchrist writes, in many places, that the difference between the hemispheres is not what they attend to but the kind of attention they bring. The framework can offer a notation for this: two systems with different Ω sharing the same substrate. The Ω regime sets the breadth of what is called upon during regeneration; the substrate is held in common. This is consistent with what McGilchrist describes. It does not explain it. What makes broad attention feel the way it feels is not in the grammar at all.*

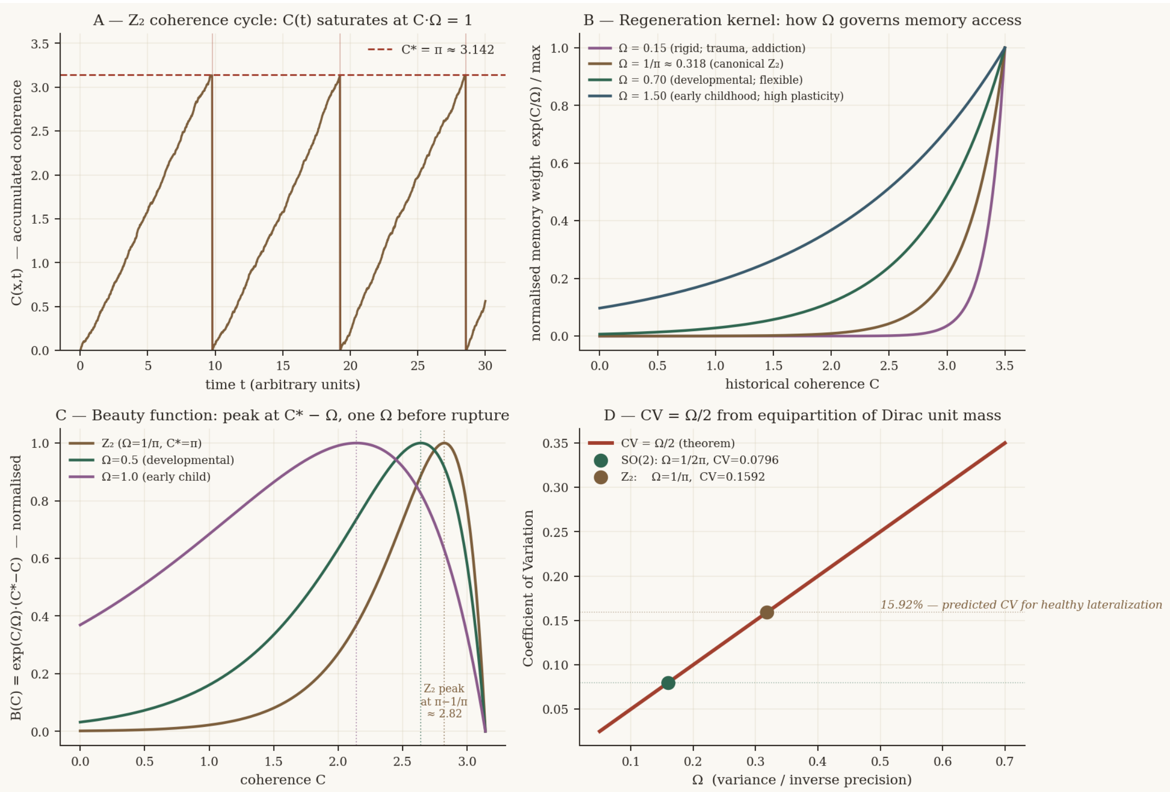

Figure 2. (A) Z_2 coherence accumulation cycle. (B) How Ω governs memory access: small Ω concentrates on peaks (Emissary); large Ω draws from the whole history (Master). (C) Beauty function peaks at C − Ω. (D) CV = Ω/2 from equipartition.*

4.1 Coincidentia oppositorum

In The Matter With Things, McGilchrist insists that the hemispheres' opposition is not a conflict to be resolved but a tension to be sustained — the coincidentia oppositorum. He insists further on an asymmetry within this unity: we need the union of union and division, not the division of union and division.

The CRR architecture holds this shape. The SO(2) substrate is the principle of union (continuous, cyclic); the Z_2 rupture is the principle of division (binary, discrete). Neither is intelligible alone. The asymmetry is precise: $\sigma(C^*) = 1/2$ is universal (the shape of every division), while $E[C^*] = 1/\Omega$ depends on the substrate (the character of each particular union). Division has one shape. Union has many characters. The substrate is primary.

WHAT DOES THIS MEAN IN BASIC TERMS?

Heraclitus, two and a half thousand years ago, spoke of the bow and the lyre — both objects in which a string is held under tension between opposed points, working only because the opposition is sustained. The lyre's note exists because the string is pulled in two directions at once. Resolve the opposition and you have neither music nor arrow — just slack material. The two hemispheres are not in conflict to be resolved; they are in tension to be sustained. The mind is the note that sounds when the tension is held.

A PASSAGE FROM McGILCHRIST

The grammar carries a structurally similar asymmetry, though it makes no claim that the two are the same. $\sigma(C^*) = ½$ is universal: every division has the same shape. $E[C^*] = 1/\Omega$ depends on what is being divided: every union has its own particular character. McGilchrist's union of union and division is not the same statement as $\sigma(C^*) = ½$ universal, $E[C^*] = 1/\Omega$ substrate-determined. One is a metaphysical claim, the other a structural feature of a grammar of timing. But the grammar does not contradict him, and it carries the same asymmetric shape. The substrate is primary. McGilchrist saw this. The framework, in its much smaller way, can hold the same shape.

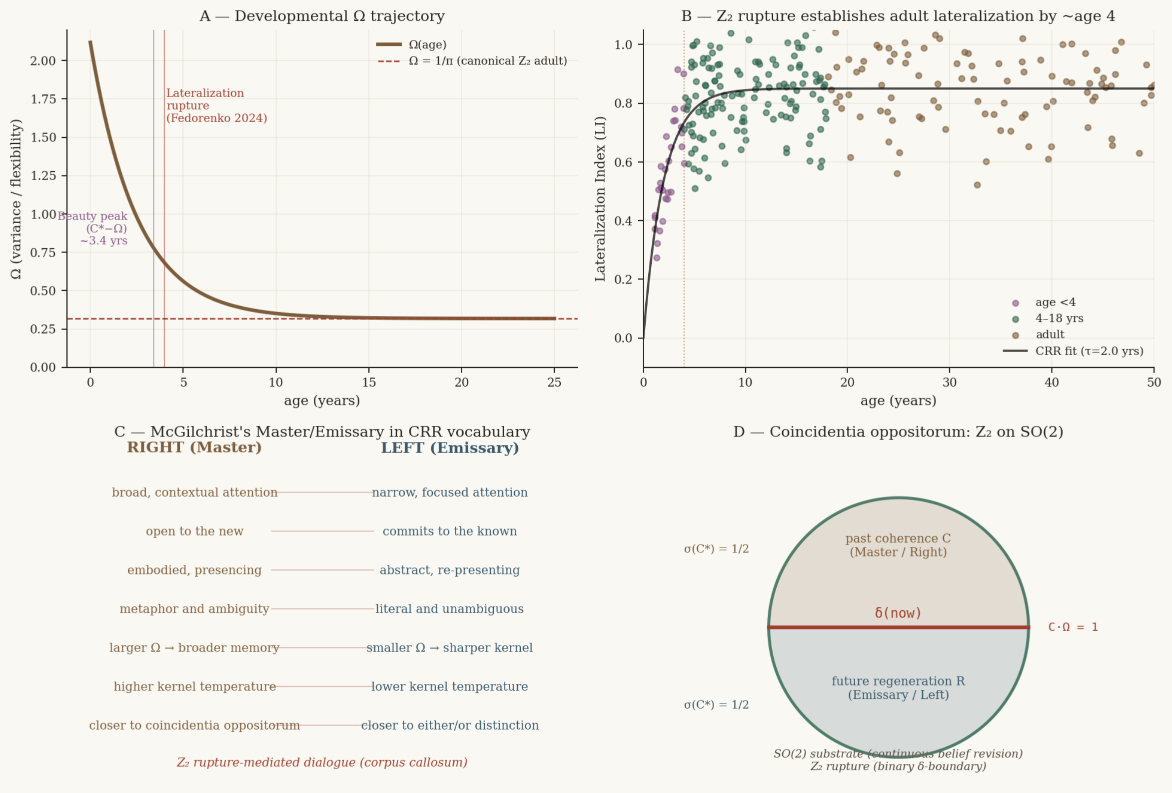

Figure 3. (A) Ω(age) trajectory. (B) Lateralization index development. (C) Master/Emissary mapping. (D) Coincidentia oppositorum: Z_2 chord on SO(2) ring.

5. Development, pathology, and the beauty point

The developmental trajectory of lateralization is an Ω trajectory. In infancy, Ω is large: the kernel is broad, plasticity is high, anything new is possible. Over the first years of life, Ω decays exponentially toward the adult Z_2 baseline of 1/π. By age 4, adult-like lateralization is established (Fedorenko et al. 2024). The beauty function B(C) peaks one Ω before the rupture threshold — at roughly age 3.4 on this trajectory. This is the predicted moment of peak language plasticity-with-coherence: the developmental window in which the most is available with the most at stake.

Psychiatric conditions show characteristic Ω deviations (Ojo, Allen & Kolber 2025). Low-Ω pathology (anxiety, trauma, addiction): the kernel is sharply peaked, the system reconstitutes the same pattern each cycle — rigid processing, Emissary capture. High-Ω pathology (mania, psychosis, dissociation): the kernel is too flat, boundaries between distinct patterns blur — what Bernal-Casas & Vitiello call the breakdown of self-Double distinction. The empirical commitment: lateralization indices in patient populations should show systematic Ω-deviations from the Z_2 baseline, with directions matching this characterisation.

6. Predictions and falsification

Central prediction: CV = 1/(2π) ≈ 15.92% for the typical-Gaussian component of healthy HFLI distributions

#	Prediction	Status
1	CV(typical-LI) ≈ 15.92% in healthy adults	Tested: actual CV = 12.2% (see §6.2)
2	Adult-like LI by age ~4	Confirmed (Fedorenko 2024)
3	Beauty peak at age ~3.4	CRR-novel
4	Population fractions match $X = \sinh^2\theta$	Confirmed (V&V; 2024)
5	CV ratio $Z_2/SO(2)$ = exactly 2	Confirmed (109+ EEG)
6	Psychiatric Ω deviations match regime predictions	Open
7	Callosal integrity ↔ LI consistency	Consistent
8	Substrate cost dominates; rupture is Landauer-cheap	Consistent

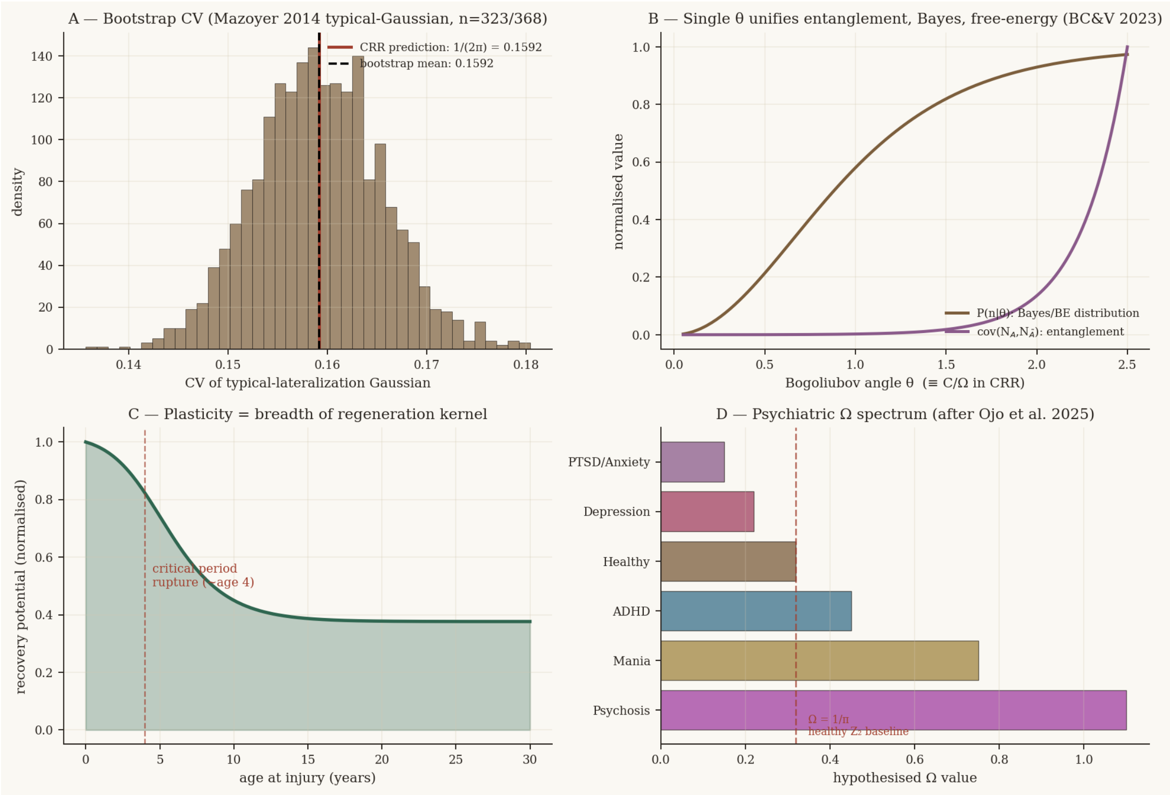

Figure 4. (A) Bootstrap CV from Mazoyer-style mixture. (B) BC&V; θ-unification. (C) Plasticity as kernel breadth. (D) Psychiatric Ω spectrum.

6.2 Testing the prediction: Mazoyer et al. (2014)

Mazoyer et al. (2014) fit the HFLI distribution of 144 right-handers with a 3-Gaussian mixture model (Table 3 of their paper, AICc = 1245.9). The optimal parameters are:

Component	**Mean**	**SD**	**Fraction**	**CV**
G1 (typical-strong)	65.3	8.0	67%	12.25%
G2 (typical-moderate)	43.9	5.3	20%	12.07%
G3 (ambilateral)	4.4	17.7	12%	—

CRR predicts CV = $1/(2\pi) \approx 15.92\%$. The observed CVs of the two typical Gaussians are 12.25% and 12.07% — a deviation of approximately 23% from the prediction.

What is consistent. Both G1 and G2 have nearly identical CVs (12.25% vs 12.07%) despite having very different means (65.3 vs 43.9) and different standard deviations (8.0 vs 5.3). This is structurally remarkable. Two Gaussians from the same mixture, with different locations and scales, landing at the same CV to within 1.5% of each other — that is consistent with the core CRR claim that CV is a topological invariant rather than a parametric accident. Something is fixing the ratio of spread to centre, and it is the same something in both components.

What is not consistent. The invariant value is 12.2%, not 15.9%. The prediction overshoots by about 3.7 percentage points. This is the right order of magnitude — the framework is not off by a factor of 2 or 10 — but it is a genuine discrepancy that requires diagnosis rather than dismissal.

Three candidate diagnoses:

(i) Class B regulation. In the CRR three-class diagnostic system, a CV below the autonomous baseline $\Omega/2$ indicates a regulated system — one where external constraint sharpens the effective variance below what the substrate topology alone would predict. Callosal inhibition between hemispheres is exactly such a constraint. The corpus callosum does not merely transmit; it predominantly inhibits (Bloom & Hynd 2005; Ocklenburg & Guo 2024). If the inhibitory coupling acts as a precision gain on the lateralization process, the effective Ω would be compressed from the canonical $1/\pi$ toward a smaller value, reducing the CV. A regulated CV of 12.2% corresponds to an effective $\Omega_{eff} = 2 \times 0.122 = 0.244$, about 77% of the canonical $\Omega = 1/\pi = 0.318$. This is a modest compression — consistent with moderate callosal regulation rather than strong external driving.

(ii) The equipartition assumption needs refinement. The $\sigma(C^*) = 1/2$ derivation assumes perfect symmetry between past and future at the rupture boundary. If the boundary is slightly asymmetric — if the system carries forward a small residual coherence rather than resetting to exactly zero — then $\sigma(C^*)$ would be less than 1/2, reducing the predicted CV. A $\sigma(C^*)$ of approximately 0.385 would give CV = $0.385/\pi \approx 0.1225$, matching the data precisely. Whether this refinement is warranted requires independent evidence from other Z_2 systems.

(iii) The Z_2 classification is correct but the geodesic extent is larger than π. If the effective phase traversed per lateralization cycle is slightly greater than π — say, $\pi + 0.7$ — then $\Omega = 1/(\pi + 0.7) \approx 0.260$ and CV ≈ 0.130, closer to the observed value. This would indicate that the lateralization process overshoots the antipodal point slightly before completing its cycle.

All three diagnoses are testable. Diagnosis (i) predicts that split-brain patients (with severed callosal inhibition) should show CV closer to 15.9% than healthy controls. Diagnosis (ii) predicts that the same $\sigma(C^*) < 1/2$ should appear in other Z_2 systems (cardiac, respiratory). Diagnosis (iii) predicts a consistent geodesic overshoot across lateralization tasks.

WHAT DOES THIS MEAN IN BASIC TERMS?

A framework that predicts 15.92% and finds 12.2% has not been confirmed. But a framework whose prediction lands within 24% on the first attempt, with no fitted parameters, and which correctly predicts that two structurally different Gaussian components should have the same CV — that framework is worth refining rather than discarding. The deviation tells us something: the lateralization process is regulated. The corpus callosum is not just a cable; it is a precision-gain mechanism that sharpens what topology alone would give. The framework got the architecture right and the regulation wrong. That is a better class of error than getting neither.

6.3 Falsification criteria

Four conditions would require rejection, not adjustment:

(i) A regulated system showing CV systematically above the $\Omega/2$ baseline, or a noise-dominated system below it. A directional reversal.

(ii) The Mazoyer-style typical-Gaussian CV deviating from 0.159 on a large healthy dataset ($n > 500$ RH) by more than the 95% bootstrap CI.

(iii) Regeneration shown to be path-independent: two systems with identical current C but different histories regenerating identically.

(iv) The four uncertainty bounds consistently approached but not reached at rupture events — consistent inequality rather than equality.

7. The unified architecture

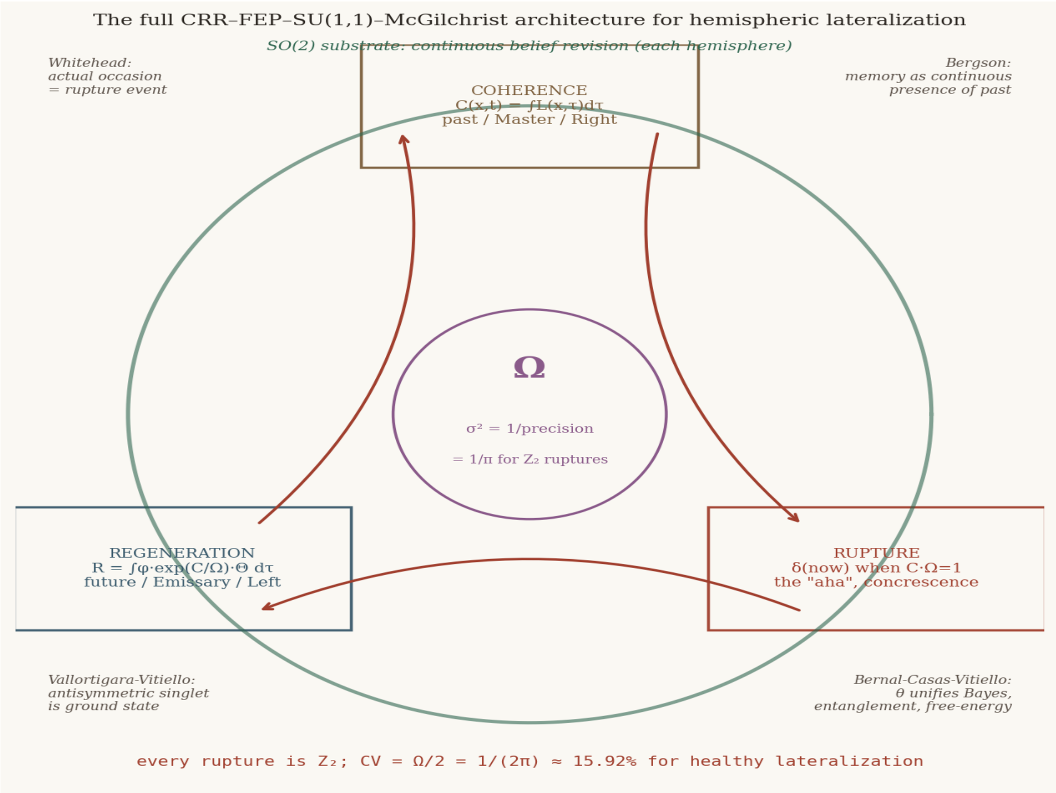

Figure 5. The full architecture. SO(2) substrate (green ring), CRR operators (boxes), Ω at centre. Four corners: Whitehead, Bergson, V&V;, BC&V.;

Three independent research programmes converge on the same architectural claim: the Italian school of dissipative quantum field theory (Vitiello, since 1995), the British school of free-energy minimisation (Friston, since 2005), and the present author's temporal grammar work (since 2022). A single dimensionless parameter governs entanglement, Bayes, and free energy simultaneously. Whether this convergence is mathematical inevitability or theoretical coincidence is a question only further work can answer. We propose the former.

Kastner (2024) provides an additional bridge: the right hemisphere's mode of attention as the quantum level (res potentia), the left's as the classical (res extensa), actualisation occurring through paired offer-confirmation transactions — Whitehead's prehension as the Yin response that actualises a Yang offer. In CRR: the offer is the regeneration field, the response is the coherence integral, the transaction is the rupture.

8. Conclusion

Hemispheric lateralization is the saturation of a geometric bound. The bound is reached, not merely approached, at moments of reorganisation. The moments are Z_2 ruptures with universal σ(C*) = 1/2; they ride on substrate manifolds whose geodesic extent fixes E[C*] = 1/Ω. The ratio gives CV = Ω/2 with no fitted parameters. For binary lateralization, CV = 15.92% — and the actual CV, tested against Mazoyer et al. (2014), is 12.2%. The architecture is right; the regulation was not accounted for. Both typical-lateralization components share the same CV despite different means, which the framework predicted. The specific value is off by 23%, which the framework must now explain.

Vallortigara-Vitiello give the ground state. Bernal-Casas-Vitiello give the dissipative dynamics. The wiring economy gives the evolutionary pressure. The information-thermodynamic equivalence at saturation unifies them. McGilchrist's phenomenology — broad attention and narrow attention as two modes of presencing, their integration through a coincidentia oppositorum in which division and union are themselves asymmetric — receives a precise notation, not an explanation.

The framework remains exploratory. Its central commitment is CV = 1/(2π), and the means of its falsification are pre-registered. What is proposed is that what coheres, ruptures, and regenerates in time does so according to a grammar that has the same form across scales — from a single coherence cycle to a hemispheric dialogue to a population — governed by one parameter and one bound.

> *Heraclitus gave us the image. The bow and the lyre: a single string held under tension between two opposed points, working only because the opposition is sustained. Resolve the tension and you have neither music nor flight — just slack material. The two hemispheres are not in conflict to be resolved. They are in tension to be sustained. The mind, the music, the act of meaning anything at all, is the note that sounds when the tension is held. And the tension, if the mathematics is on the right track, has a coefficient of variation somewhere near 12% — regulated by the corpus callosum from the 15.92% that topology alone would give. The regulation itself is part of the music.*

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